

MPPG 2a

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TABLE 1. Recommended minimum practices for commissioning and QA of an IGRT system.

<i>Acceptance Testing and Commissioning</i>	
<i>Procedure</i>	
Customer acceptance procedures	
TPS integration	
OIS integration	
Establish routine QA baselines	
QA documentation	
<i>Routine Quality Assurance</i>	
<i>Procedure</i>	<i>Tolerance</i>
<i>Daily</i>	
<i>Safety/interlocks</i>	
<i>Functional</i>	
Imaging-treatment isocenter coincidence (SRS only)	1 mm
Positioning/repositioning (SRS only)	1 mm
Imaging-treatment isocenter coincidence (SBRT only)	2 mm
Positioning/repositioning (SBRT only)	2 mm
<i>Weekly</i>	
Imaging-treatment isocenter coincidence (non-SRS/SBRT)	2 mm
Positioning/repositioning (non-SRS/SBRT)	2 mm
<i>Semi-annually</i>	
Image scaling	2 mm
<i>Annually</i>	
<i>Imaging dose</i>	
2D MV	± 1 cGy of baseline value
2D kV (static imaging mode)	± 3 mGy of baseline value
2D kV (fluoroscopy mode)	± 1 cGy/min of baseline value
All 3D imaging modes	± 1 cGy of baseline value
<i>Image quality</i>	
2D (spatial resolution, contrast)	Baseline value
3D (uniformity, spatial resolution, contrast)	
<i>Upgrade/Repair/Service</i>	
Verify / Reestablish QA baselines (as appropriate)	

SRS = stereotactic radiosurgery; SBRT = stereotactic body radiation therapy.